The Road Not Taken: College Board Geomarkets and Recruiting Visits by Selective Colleges

1 Research questions

Research interest: performativity (MacKenzie and Millo 2003) in the relationship between College Board Geomarkets, the Market Segment Model, and patterns of recruiting visits to high schools.

1. Do College Board Geomarkets (X) explain which high schools receive recruiting visits (Y) from public research and selective private institutions?
2. To what extent are visit patterns consistent with recommendations from the Market Segment Model?
3. Does Geomarket popularity (Z) mediate the relationship between school composition (X; %free reduced lunch; %Black/Hispanic) and probability of a high school getting a visit (Y)?

2 Front end

2.1 Dependence and Patronage in the Enrollment Economy

- Dependence (Pfeffer and Salancik 1978) and patronage (Stevens and Gebre-Medhin 2016)
- "Enrollment economy" from Clark (1956) and Winston (1999); sketch enrollment economy for sample
  - 12 selective private liberal arts colleges; 14 selective private research universities
  - 16 public research universities; from "national service" patronage to reliance on same enrollment economy as selective privates (Jaquette and Curs 2015; Jaquette et al. 2024)
- Enrollment management (EM) is responsible for stable flow of resources (students); colleges depend on high schools – dyadic relationships – to provide these students
- Scholarship on recruiting that focuses on linkages between colleges and high schools
  - Coopt dependencies that cannot be acquired/substituted (Stevens 2007; Khan 2010)
  - Scholarship on recruiting (Salazar et al. 2021; Salazar 2022; Jaquette et al. 2024) incorrectly assumes that EM behavior is a function of *individual* colleges, conditioned by macro logics

2.2 The Case: Geomarkets and the Market Segment Model

- Geomarkets and Market Segment Model developed by Zemsky and Oedel (1983)
- Origins in concerns about demographics; involve membership orgs COFHE + College Board
- Categorize students into four "market segments" based on SAT score-sending; creation of Geomarkets
- Market Segment Model: social class determines demand for specific college located in specific place
  - Class composition varies by Geomarket; implications for which Geomarkets to recruit from

2.3 Market Devices, Commensuration, and Classification

Introduce salient concepts/literatures and pepper-in additional context about the case

- *Market devices* (Muniesa et al. 2007; MacKenzie and Millo 2003; Krippner 2024)
  - Market devices useful for shifting attentiopn to individual components – techniques, theories, cultural logics, histories – foundational to products and markets
  - Geomarkets and Market Segment Model are market devices that build on SAT
  - Performativity; options prices began behaving like Black-Scholes model because logic of model built into trading software (MacKenzie and Millo 2003);
    - Similarly, logic of Market Segment Model – a class-based theory of geography of student demand – built into College Board Enrollment Planning Service (EPS) software
- *Commensuration* (Espeland and Stevens 1998; Fourcade 2011)
  - Analysis of UK school "league tables" (McArthur and Reeves 2022); commensuration as a mechanism of social reproduction
  - SAT as a market device that facilitates commensuration (Hoxby 2009);
  - Geomarkets are a commensurating market device that enables colleges to compare geographic localities based on class composition and number of students from each market segment
- *Ordinalizaiton and classificaiton situtions* (Fourcade and Healy 2013, 2024, 2017; Fourcade 2016)
  - Fourcade (2016) defines three kinds of classification; ascendance of ordinal classifications
  - Classification situations (Fourcade and Healy 2013)
    - goal of classification situtations is "good matches" between ordinal hierarchy of products and ordinal hierarchy of consumers
    - Market Segment Model as ordinal market device that amplifies inequality; class-based hierarchy of students matched to class-based hierarchy of colleges
- Hook; most scholarship on quantification in sociology of education focuses on third-parties that classify producers/sellers on behalf of consumers (e.g., college/school rankings)
  - Scholarship shows how orgs (Ray 2019), schools (Domina et al. 2017) reproduce inequality
  - Empirical scholarship in sociology of education fails to consider third-party intermediaries – along the lines of classification situations – that classify students on behalf of colleges

2.4 RQs and Expected Results

Expected results speak to performativity consequences of Geomarkets and Market Segment Model

- **RQ1**: Do college board Geomarkets explain which high schools receive recruiting visits from public research and selective private PSIs?
  - Expect Geomarkets explain recruiting visits, even after strong controls, because Geomarkets/Market Segment Model incorporated into EPS software and Geomarkets incorporated into CRMs – Slate, Scoir, PeopleSoft – admissions offices use to assign recruiters to territories
  - Table 1 shows model fit stats for models of visits to school i by college j; takeaway: models with Fixed Effects for Geomarket explain more variation
  - Appendix Table A1 shows regression coefficients (visits to public HS only)
  - Appendix Table A2 shows model fit stats for models of visits to school i, separate models by college j (visits to public HS only)
- **RQ2**: To what extent are visit patterns consistent with recommendations from the Market Segment?
  - We expect that recruiting visit patterns of most colleges are consistent with Market Segment Model recommendations of **(A)** a regional/national recruiting strategy that **(B)** focuses on high schools in affluent Geomarkets
  - ***RQ2A***: Figure 1 shows geography of 2016 freshman enrollment for sample; Figure 2 shows 2017 recruiting by local, in-state, regional, national markets; Figure 3 shows 2017 recruiting by EPS region (New England; Middle States; Midwest; South; Southwest; West)
  - ***RQ2B***: Table 2 and Table 3, respectively, show top-30 Geomarkets for visits to public schools and private schools; Figure 4 shows Lorenz-style concentration curves of recruiting visits by Geomarket affluence
- **RQ3**: Does geomarket popularity (Z) mediate the relationship between school composition (X; % free reduced lunch; % Black/Hispanic) and probability of a high school getting a visit (Y)?
  - We expect that high SES schools and low %Black/Hispanic schools will benefit more from being in a popular Geomarket because these schools are a "good match" with popular Geomarket whereas poor/minority schools are a bad match. Fourcade and Healy (2017, 17): "'good matches' are perceived as natural because they fit well with how things already are."
  - Table 4 shows fixed effects regression model with interaction between Geomarket popularity and school SES/racial composition

3 References

Clark, Burton R. 1956. "Organizational Adaptation and Precarious Values: A Case Study." *American Sociological Review* 21 (3): 327–36. <Go to ISI>://A1956CAX7800009.

Domina, Thurston, Andrew Penner, and Emily Penner. 2017. "Categorical Inequality: Schools as Sorting Machines." *Annual Review of Sociology* 43: 311–30. https://doi.org/10.1146/annurev-soc-060116-053354.

Espeland, Wendy N., and Mitchell L. Stevens. 1998. "Commensuration as a Social Process." *Annual Review of Sociology* 24: 313–43. https://doi.org/10.1146/annurev.soc.24.1.313.

Fourcade, Marion. 2011. "Cents and Sensibility: Economic Valuation and the Nature of "Nature"." *American Journal of Sociology* 116 (6): 1721–77. https://doi.org/10.1086/659640.

Fourcade, Marion. 2016. "Ordinalization: Lewis a. Coser Memorial Award for Theoretical Agenda Setting 2014." *Sociological Theory* 34 (3): 175–95. https://doi.org/10.1177/0735275116665876.

Fourcade, Marion, and Kieran Joseph Healy. 2013. "Classification Situations: Life-Chances in the Neoliberal Era." Journal Article. *Accounting Organizations and Society* 38 (8): 559–72. https://doi.org/10.1016/j.aos.2013.11.002.

Fourcade, Marion, and Kieran Joseph Healy. 2017. "Seeing Like a Market." *Socio-Economic Review* 15 (1): 9–29. https://doi.org/10.1093/ser/mww033.

Fourcade, Marion, and Kieran Joseph Healy. 2024. *The Ordinal Society*. Harvard University Press.

Hoxby, Caroline M. 2009. "The Changing Selectivity of American Colleges." *Journal of Economic Perspectives* 23 (4): 95–118. https://doi.org/DOI: 10.1257/jep.23.4.95.

Jaquette, Ozan, and Bradley R. Curs. 2015. "Creating the Out-of-State University: Do Public Universities Increase Nonresident Freshman Enrollment in Response to Declining State Appropriations?" *Research in Higher Education* 56 (6): 535–65. https://doi.org/10.1007/s11162-015-9362-2.

Jaquette, Ozan, Crystal Han, and Irma Castañeda. 2024. "The Private School Network: Recruiting Visits to Private High Schools by Public and Private Universities." Journal Article. *Research in Higher Education* 65 (6): 1269–315. https://doi.org/10.1007/s11162-024-09783-w.

Khan, Shamus Rahman. 2010. "Getting in: How Elite Schools Play the College Game." In *Educating Elites Class Privilege and Educational Advantage*, edited by Adam Howard and Rubén A. Gaztambide-Fernandez. Rowman & Littlefield. https://ebookcentral.proquest.com/lib/umichigan/detail.action?docID=634205.

Krippner, Greta R. 2024. "Gendered Market Devices: The Persistence of Gender Discrimination in Insurance Markets." *American Journal of Sociology* 130 (3): 595–643. https://doi.org/10.1086/732318.

MacKenzie, Donald, and Yuval Millo. 2003. "Constructing a Market, Performing Theory: The Historical Sociology of a Financial Derivatives Exchange." *American Journal of Sociology* 109 (1): 107–45. https://doi.org/10.1086/374404.

McArthur, Daniel, and Aaron Reeves. 2022. "The Unintended Consequences of Quantifying Quality: Does Ranking School Performance Shape the Geographical Concentration of Advantage?" *American Journal of Sociology* 128 (2): 515–51. https://doi.org/10.1086/722470.

Muniesa, F., Y. Millo, and M. Callon. 2007. "An Introduction to Market Devices." *Sociological Review* 55: 1–12. https://doi.org/10.1111/j.1467-954X.2007.00727.x.

Pfeffer, Jeffrey, and Gerald R. Salancik. 1978. *The External Control of Organizations: A Resource Dependence*

Perspective. Harper & Row.

Ray, V. 2019. "A Theory of Racialized Organizations." *American Sociological Review* 84 (1): 26–53. https://doi.org/10.1177/0003122418822335.

Salazar, Karina G. 2022. "Recruitment Redlining by Public Research Universities in the Los Angeles and Dallas Metropolitan Areas." *The Journal of Higher Education* 93: 585–621. https://doi.org/10.1080/00221546.2021.2004811.

Salazar, Karina G., Ozan Jaquette, and Crystal Han. 2021. "Coming Soon to a Neighborhood Near You? Off-Campus Recruiting by Public Research Universities." *American Educational Research Journal* 58 (6): 1270–314. https://doi.org/10.3102/00028312211001810.

Stevens, Mitchell L. 2007. *Creating a Class: College Admissions and the Education of Elites*. Harvard University Press.

Stevens, Mitchell L., and Ben Gebre-Medhin. 2016. "Association, Service, Market: Higher Education in American Political Development." *Annual Review of Sociology* 42: 121–42. https://doi.org/10.1146/annurev-soc-081715-074240.

Winston, G. C. 1999. "Subsidies, Hierarchy and Peers: The Awkward Economics of Higher Education." *Journal of Economic Perspectives* 13 (1): 13–36. https://doi.org/10.1257/jep.13.1.13.

Zemsky, Robert, and Penney Oedel. 1983. *The Structure of College Choice*. College Entrance Examination Board.

Table 1: Model Fit Statistics, fixed effects regression of visit to high school i by college j, separate models for all HS, public HS, private HS

term	(1) Univ × State FE	(2) Univ × EPS FE	(3) Covar + Univ FE	(4) Covar + Univ × State FE	(5) Covar + Univ × EPS FE
All High Schools					
N	1,026,984	1,026,984	1,013,166	1,013,166	1,013,166
Adj. R^2	0.097	0.168	0.136	0.210	0.258
Within R^2	0.000	0.000	0.129	0.125	0.108
RMSE	0.165	0.158	0.162	0.155	0.149
Public High Schools Only					
N	853,692	853,692	748,440	748,440	748,440
Adj. R^2	0.118	0.210	0.120	0.223	0.301
Within R^2	0.000	0.000	0.111	0.104	0.082
RMSE	0.143	0.134	0.149	0.140	0.132
Private High Schools Only					
N	173,292	173,292	167,454	167,454	167,454
Adj. R^2	0.096	0.146	0.170	0.229	0.264
Within R^2	0.000	0.000	0.153	0.147	0.137
RMSE	0.238	0.224	0.232	0.222	0.210

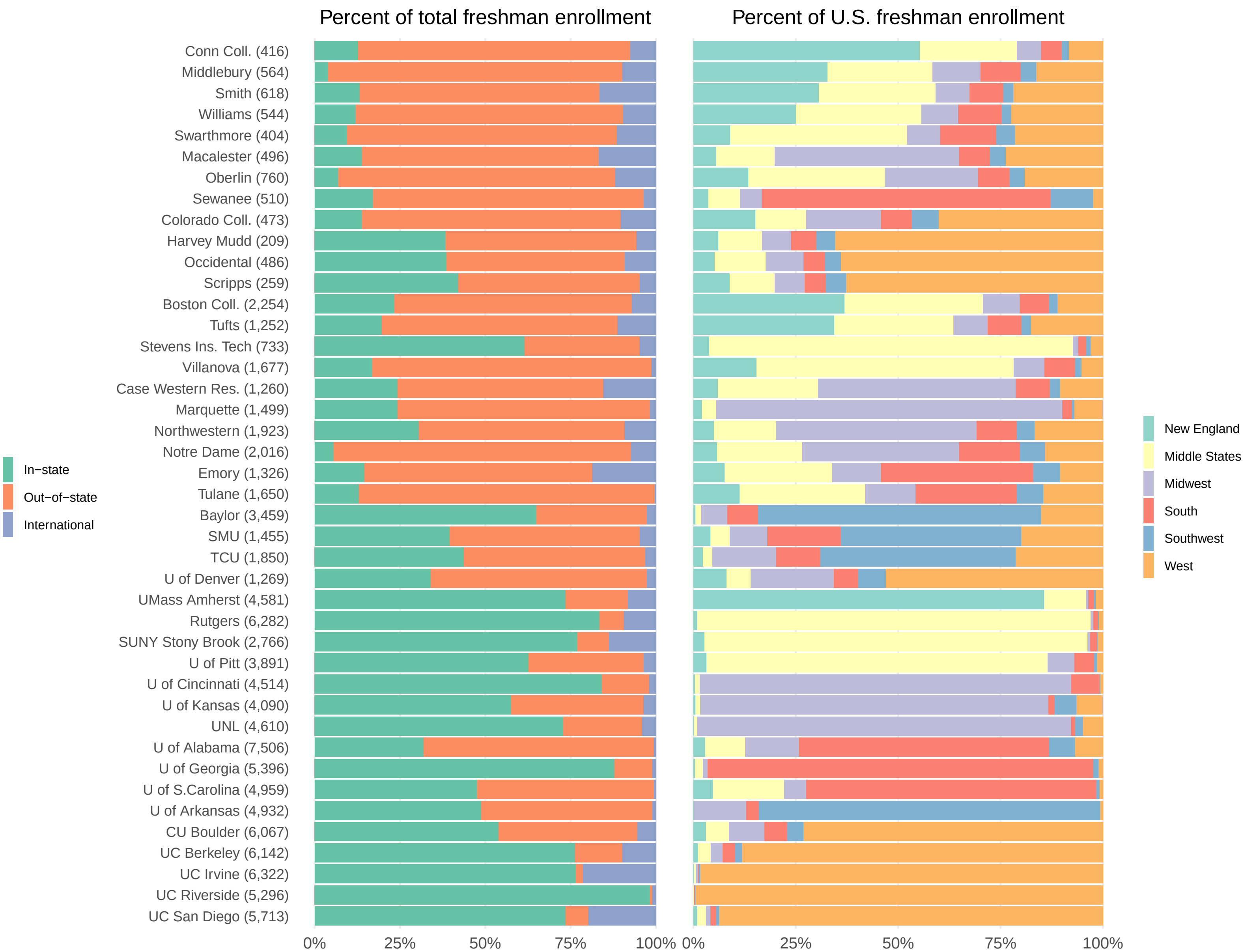

Figure 1: Geographic origin of fall 2016 freshman enrollment

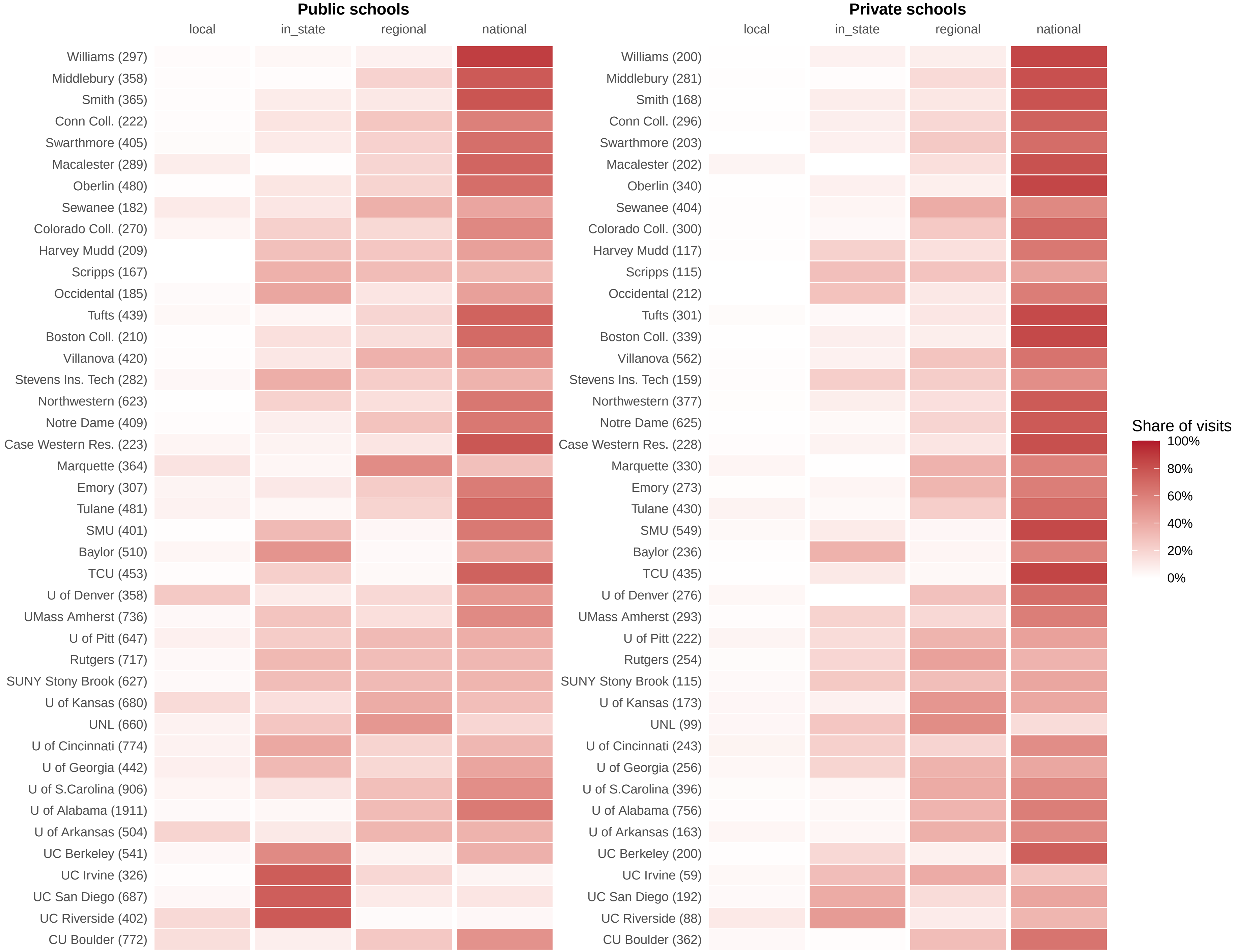

Figure 2: Recruiting visits by market segment of Geomarket

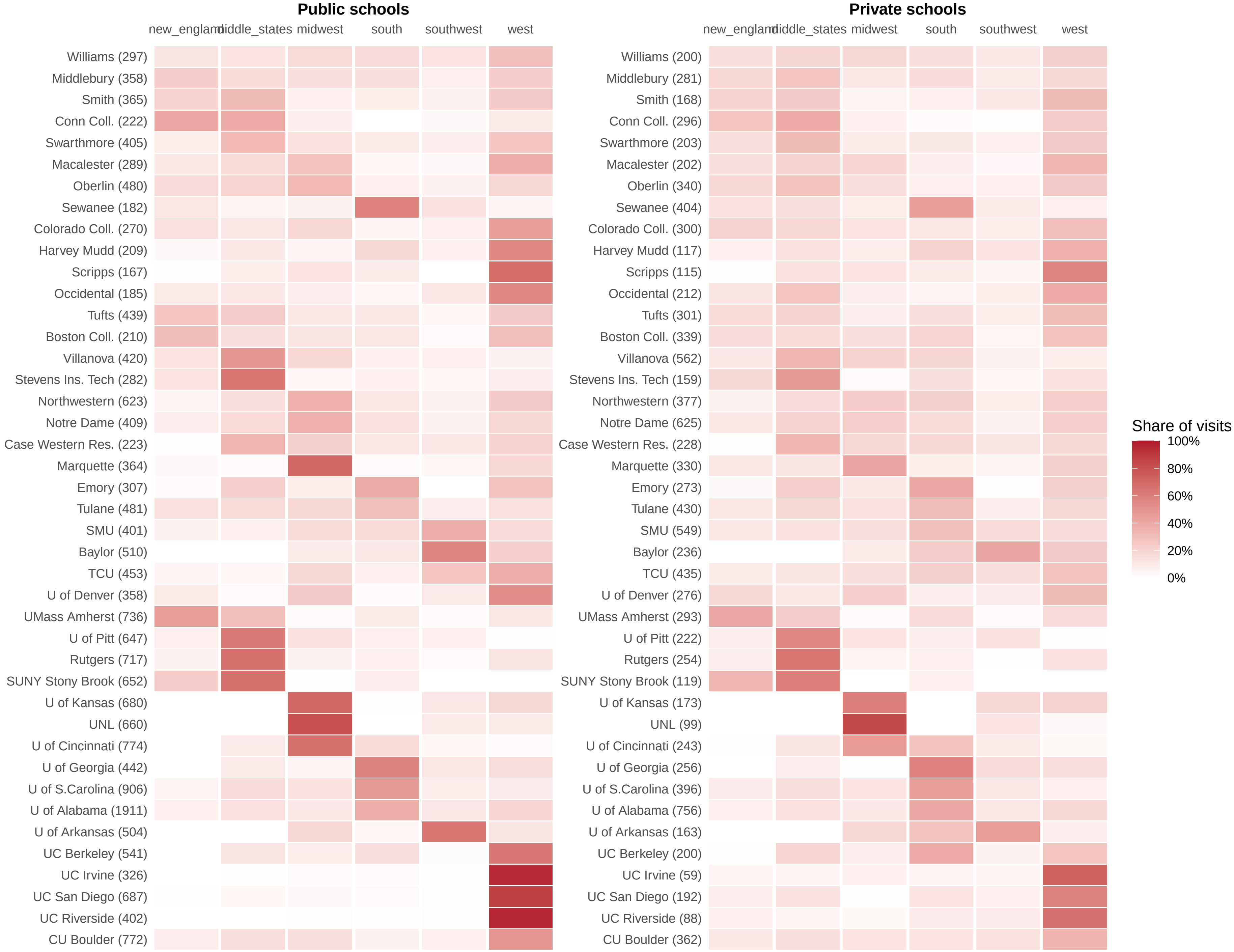

Figure 3: Recruiting visits by EPS region

Table 2: Top 30 Geomarkets ranked by visits per school (public school visits only)

Rank	EPS	Visits/Sch	MeanInc	%BA+	%Pov	%White	%Asian	%Black	%Hisp
1	IL 9 - North Shore	24.5	$259k	73.5	5.6	79.0	9.6	1.9	7.2
2	IL10 - Evanston & Skokie	17.0	$156k	58.3	8.4	64.6	17.7	5.1	9.4
3	VA 1 - Arlington & Alexandria	11.6	$175k	71.4	6.6	57.1	8.8	13.9	15.8
4	TX23 - Collin & Rockwall Co	11.2	$146k	52.3	6.0	56.8	14.5	9.4	15.6
5	IL 8 - Northwest Suburbs	11.0	$126k	44.7	6.8	58.2	15.0	3.3	20.8
6	MD 2 - Montgomery Metropolitan	10.4	$165k	59.2	6.6	43.1	14.9	18.0	19.5
7	VA 2 - Fairfax Co	10.3	$182k	62.3	5.4	50.4	19.6	9.3	16.2
8	IL12 - Western Suburbs	10.3	$134k	46.9	6.8	60.3	10.5	8.1	18.5
9	CT 3 - Fairfield Co	9.9	$160k	48.9	8.7	61.0	5.3	10.6	20.0
10	IL 7 - Chain of Lakes	9.7	$125k	39.7	8.4	57.2	7.4	7.5	25.1
11	CA11 - Santa Clara Co (excluding San Jose)	9.6	$234k	64.3	5.7	36.8	39.4	1.8	17.1
12	CO 2 - Metro Denver & Northeastern Colorado	9.4	$121k	44.8	8.7	66.2	3.8	4.3	22.1
13	CA 4 - Marin Co	9.0	$190k	60.2	6.6	70.6	5.7	2.1	16.1
14	NY17 - Northern Nassau Co	8.8	$266k	62.4	6.7	67.7	16.9	2.3	10.6
15	NJ 6 - Somerset & Mercer Co	8.8	$155k	49.3	8.5	51.8	14.6	14.8	16.4
16	CA28 - South Orange Co	8.2	$177k	56.7	7.2	57.4	19.5	1.4	16.5
17	CA17 - West Los Angeles & West Beach	8.2	$155k	60.3	11.1	50.3	11.9	9.6	22.5
18	MA10 - Milton, Lexington, & Waltham	8.1	$181k	64.0	5.8	74.2	13.1	3.4	6.0
19	GA 2 - Fulton Co	7.7	$116k	54.5	12.3	39.3	7.3	43.1	7.2
20	NY19 - Northwest Suffolk Co	7.7	$182k	47.7	4.6	76.9	5.9	3.0	12.0
21	CA 6 - Contra Costa Co	7.2	$151k	43.3	8.1	42.6	17.2	8.2	25.8
22	CA26 - Santa Ana	7.0	$131k	36.5	10.4	35.1	21.1	1.1	38.9
23	TX24 - West of Dallas/Ft. Worth Metroplex	6.9	$124k	41.3	6.9	62.6	6.9	8.9	17.9
24	NJ11 - Morris & Northern Passaic Co	6.8	$173k	53.4	5.3	71.8	9.7	2.9	13.2
25	NJ10 - Bergen Co	6.7	$159k	50.7	7.2	55.5	16.4	5.3	20.4
26	CA 8 - Alameda Co (w/o Oakland)	6.7	$165k	49.3	7.6	31.0	36.4	6.0	20.8
27	NJ 7 - Union Co	6.6	$114k	37.1	8.9	39.0	5.2	20.2	32.0
28	PA 4 - Montgomery Co	6.6	$142k	49.6	6.2	74.9	7.7	9.1	5.3
29	CA20 - South Bay	6.5	$126k	44.6	8.9	31.7	21.8	9.8	31.7
30	MA 8 - Lowell, Concord, & Wellesley	6.5	$160k	51.6	7.2	75.0	10.7	3.4	7.4

Table 3: Top 30 Geomarkets ranked by visits per school (private school visits only)

Rank	EPS	Visits/Sch	MeanInc	%BA+	%Pov	%White	%Asian	%Black	%Hisp
1	IL 9 - North Shore	31.0	$259k	73.5	5.6	79.0	9.6	1.9	7.2
2	TX19 - City of Dallas	22.4	$81k	34.9	15.6	29.8	3.4	22.9	41.7
3	IL10 - Evanston & Skokie	20.3	$156k	58.3	8.4	64.6	17.7	5.1	9.4
4	TX20 - City of Fort Worth	20.2	$91k	28.3	12.1	42.0	4.4	15.8	34.8
5	CA 7 - City of Oakland	19.5	$124k	47.0	13.4	29.6	15.7	21.7	26.3
6	IL 7 - Chain of Lakes	19.0	$125k	39.7	8.4	57.2	7.4	7.5	25.1
7	VA 1 - Arlington & Alexandria	18.8	$175k	71.4	6.6	57.1	8.8	13.9	15.8
8	CA 4 - Marin Co	17.7	$190k	60.2	6.6	70.6	5.7	2.1	16.1
9	DC 1 - District of Columbia	17.0	$155k	59.8	13.7	36.7	4.0	44.5	11.1
10	TX16 - Southwest Houston Metro Area	16.6	$106k	46.8	11.8	31.5	16.1	20.0	29.6
11	CA29 - North San Diego Co (w/o San Diego)	16.0	$144k	44.1	8.0	52.3	10.4	2.3	29.8
12	CT 4 - Waterbury & Litchfield Co	16.0	$129k	35.9	7.2	87.6	1.9	1.5	6.6
13	NH 1 - Seacoast	16.0	$138k	44.5	7.1	91.2	3.0	0.9	2.3
14	GA 2 - Fulton Co	14.9	$116k	54.5	12.3	39.3	7.3	43.1	7.2
15	CA28 - South Orange Co	14.2	$177k	56.7	7.2	57.4	19.5	1.4	16.5
16	CT 5 - Hartford & Tolland Co	14.1	$117k	39.0	11.0	63.5	5.4	11.4	16.6
17	CA31 - City of San Diego	14.0	$118k	45.4	11.4	44.7	14.7	6.8	28.9
18	MA 7 - Quincy & Plymouth Co	13.2	$137k	38.6	7.5	80.2	1.4	8.9	4.0
19	TX 6 - Austin & Central Texas	13.1	$114k	43.0	9.4	53.7	5.0	6.4	31.9
20	NY19 - Northwest Suffolk Co	13.0	$182k	47.7	4.6	76.9	5.9	3.0	12.0
21	OH 5 - Cuyahoga, Geauga, & Lake Co	12.8	$106k	36.4	7.1	87.2	2.3	4.6	3.5
22	OR 1 - Greater Portland (West)	12.8	$130k	53.6	9.4	69.6	9.0	3.3	12.1
23	MO 1 - Kansas City & St. Joseph	12.6	$84k	28.7	11.9	76.0	1.5	11.9	6.7
24	NJ 6 - Somerset & Mercer Co	12.2	$155k	49.3	8.5	51.8	14.6	14.8	16.4
25	CA15 - San Fernando Valley (East)	12.1	$123k	51.6	12.0	58.4	6.8	5.1	24.9
26	CO 2 - Metro Denver & Northeastern Colorado	12.0	$121k	44.8	8.7	66.2	3.8	4.3	22.1
27	MD 2 - Montgomery Metropolitan	11.9	$165k	59.2	6.6	43.1	14.9	18.0	19.5
28	CA11 - Santa Clara Co (excluding San Jose)	11.6	$234k	64.3	5.7	36.8	39.4	1.8	17.1
29	CA26 - Santa Ana	11.5	$131k	36.5	10.4	35.1	21.1	1.1	38.9
30	CT 2 - New Haven & Middlesex Co	11.3	$118k	37.1	10.8	65.3	3.9	11.3	16.6

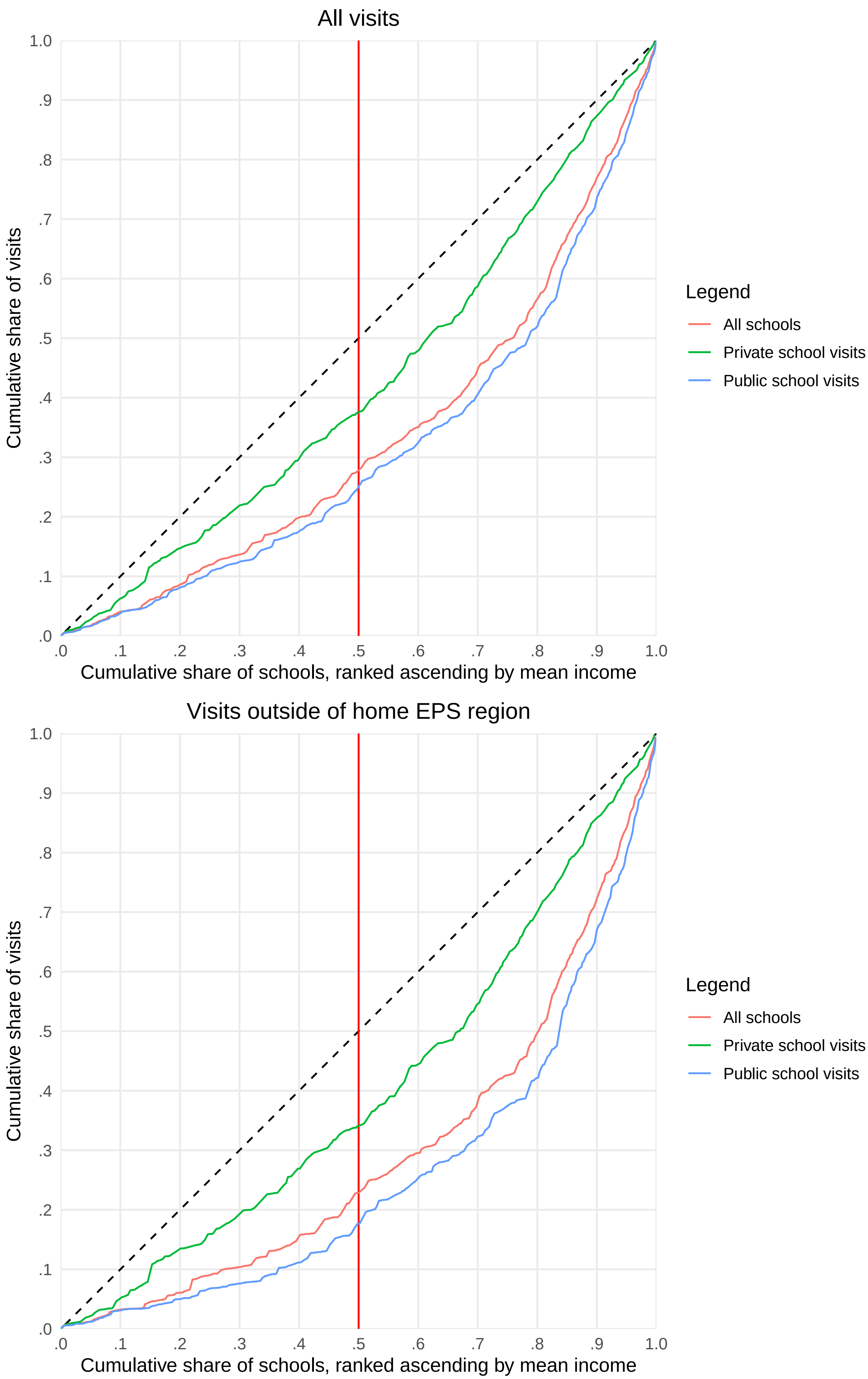

Figure 4: Lorenz-style concentration curves of recruiting visits by Geomarket affluence

Table 4: Fixed Effects Regression Model with Interaction between Geomarket Popularity and School SES/Racial Composition

	Model		Interaction Free-Reduced Lunch		Interaction Black/Hisp	
	Estimate	Std Error	Estimate	Std Error	Estimate	Std Error
hs_pct_free_reduced_lunch_decileD2	-0.008*	0.003	0.016***	0.003	-0.007*	0.003
hs_pct_free_reduced_lunch_decileD3	-0.006	0.004	0.025***	0.004	-0.004	0.003
hs_pct_free_reduced_lunch_decileD4	-0.005	0.004	0.030***	0.004	-0.003	0.004
hs_pct_free_reduced_lunch_decileD5	-0.003	0.004	0.034***	0.004	-0.001	0.004
hs_pct_free_reduced_lunch_decileD6	-0.004	0.004	0.033***	0.004	-0.001	0.004
hs_pct_free_reduced_lunch_decileD7	-0.004	0.004	0.035***	0.004	-0.001	0.004
hs_pct_free_reduced_lunch_decileD8	-0.002	0.004	0.036***	0.005	0.001	0.004
hs_pct_free_reduced_lunch_decileD9	-0.001	0.005	0.038***	0.005	0.002	0.004
hs_pct_free_reduced_lunch_decileD10	-0.000	0.004	0.043***	0.004	0.002	0.004
hs_pct_bl_hisp_nat_decileD1	0.005	0.002	0.006*	0.002	0.020***	0.003
hs_pct_bl_hisp_nat_decileD2	0.003	0.002	0.004	0.002	0.012***	0.003
hs_pct_bl_hisp_nat_decileD3	0.003	0.003	0.003	0.002	0.006*	0.002
hs_pct_bl_hisp_nat_decileD4	0.002	0.002	0.001	0.002	0.002	0.002
hs_pct_bl_hisp_nat_decileD6	-0.005***	0.001	-0.004**	0.001	-0.002	0.002
hs_pct_bl_hisp_nat_decileD7	-0.004*	0.002	-0.001	0.002	0.001	0.003
hs_pct_bl_hisp_nat_decileD8	-0.007*	0.003	-0.002	0.002	0.004	0.003
hs_pct_bl_hisp_nat_decileD9	-0.011*	0.004	-0.003	0.004	0.006	0.003
hs_pct_bl_hisp_nat_decileD10	-0.016**	0.005	-0.005	0.005	0.012**	0.004
peps_n_vis01_per_sch_all	0.504***	0.035	0.955***	0.049	0.695***	0.046
hs_pct_free_reduced_lunch_decileD2:peps_n_vis01_per_sch_all			-0.327***	0.032		
hs_pct_free_reduced_lunch_decileD3:peps_n_vis01_per_sch_all			-0.480***	0.041		
hs_pct_free_reduced_lunch_decileD4:peps_n_vis01_per_sch_all			-0.569***	0.047		
hs_pct_free_reduced_lunch_decileD5:peps_n_vis01_per_sch_all			-0.629***	0.048		
hs_pct_free_reduced_lunch_decileD6:peps_n_vis01_per_sch_all			-0.609***	0.047		
hs_pct_free_reduced_lunch_decileD7:peps_n_vis01_per_sch_all			-0.675***	0.043		
hs_pct_free_reduced_lunch_decileD8:peps_n_vis01_per_sch_all			-0.665***	0.059		
hs_pct_free_reduced_lunch_decileD9:peps_n_vis01_per_sch_all			-0.697***	0.071		
hs_pct_free_reduced_lunch_decileD10:peps_n_vis01_per_sch_all			-0.813***	0.039		
hs_pct_bl_hisp_nat_decileD1:peps_n_vis01_per_sch_all					-0.456***	0.067
hs_pct_bl_hisp_nat_decileD2:peps_n_vis01_per_sch_all					-0.200**	0.058
hs_pct_bl_hisp_nat_decileD3:peps_n_vis01_per_sch_all					-0.067	0.059
hs_pct_bl_hisp_nat_decileD4:peps_n_vis01_per_sch_all					-0.005	0.040
hs_pct_bl_hisp_nat_decileD6:peps_n_vis01_per_sch_all					-0.089**	0.028
hs_pct_bl_hisp_nat_decileD7:peps_n_vis01_per_sch_all					-0.128***	0.035
hs_pct_bl_hisp_nat_decileD8:peps_n_vis01_per_sch_all					-0.223***	0.042
hs_pct_bl_hisp_nat_decileD9:peps_n_vis01_per_sch_all					-0.316***	0.052
hs_pct_bl_hisp_nat_decileD10:peps_n_vis01_per_sch_all					-0.487***	0.058
N	678,804		678,804		678,804	
Adj. R^2	0.272		0.298		0.282	
Within R^2	0.151		0.181		0.162	
RMSE	0.140		0.138		0.139	

Significance codes: *** p<0.001, ** p<0.01, * p<0.05

4 Appendix

Table A1: Fixed effects regression models of visit to high school i by college j (public high schools only)

term	(1) Univ × State FE	(2) Univ × EPS FE	(3) Covar + Univ FE	(4) Covar + Univ × State FE	(5) Covar + Univ × EPS FE
hs_g11			0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
hs_pct_asian			0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
hs_pct_black			-0.000*** (0.000)	-0.000* (0.000)	-0.000 (0.000)
hs_pct_hispanic			-0.000 (0.000)	-0.000* (0.000)	-0.000** (0.000)
hs_pct_amerindian			-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
hs_pct_nativehawaii			0.001** (0.000)	-0.001** (0.000)	-0.001*** (0.000)
hs_pct_tworaces			0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
hs_overall_niche_letter_gradeA+			0.126*** (0.013)	0.128*** (0.013)	0.120*** (0.011)
hs_overall_niche_letter_gradeA			0.027*** (0.006)	0.028*** (0.005)	0.027*** (0.005)
hs_overall_niche_letter_gradeA-			0.004 (0.003)	0.005 (0.002)	0.005 (0.003)
hs_overall_niche_letter_gradeB+			0.000 (0.002)	-0.000 (0.002)	0.000 (0.002)
hs_overall_niche_letter_gradeB			-0.002 (0.001)	-0.002 (0.002)	-0.002 (0.002)
hs_overall_niche_letter_gradeB-			-0.004** (0.001)	-0.004* (0.002)	-0.003* (0.002)
hs_overall_niche_letter_gradeC+			-0.004** (0.001)	-0.004* (0.002)	-0.003 (0.002)
hs_overall_niche_letter_gradeC			-0.003* (0.001)	-0.004* (0.001)	-0.003 (0.002)
hs_overall_niche_letter_gradeC-			-0.003* (0.001)	-0.005** (0.001)	-0.004* (0.002)
hs_magnet011			-0.006 (0.003)	-0.002 (0.003)	-0.000 (0.003)
hs_school_typecareer and technical school			-0.029** (0.009)	-0.034** (0.010)	-0.030** (0.009)
hs_school_typealternative education school			-0.005* (0.002)	0.003 (0.003)	0.004 (0.003)
hs_pct_free_reduced_lunch			-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)

(continued)

term	(1) Univ × State FE	(2) Univ × EPS FE	(3) Covar + Univ FE	(4) Covar + Univ × State FE	(5) Covar + Univ × EPS FE
hs__pct__prof__math			-0.020**	0.002	0.004
			(0.007)	(0.006)	(0.005)
hs__pct__prof__rla			0.006	0.000	0.001
			(0.007)	(0.005)	(0.005)
hs__zip__inc__house__mean__decileD2			0.002	0.000	0.000
			(0.001)	(0.001)	(0.001)
hs__zip__inc__house__mean__decileD3			0.002	-0.001	-0.001
			(0.002)	(0.001)	(0.001)
hs__zip__inc__house__mean__decileD4			0.003	0.000	0.000
			(0.002)	(0.002)	(0.002)
hs__zip__inc__house__mean__decileD5			0.003	-0.001	-0.001
			(0.003)	(0.002)	(0.002)
hs__zip__inc__house__mean__decileD6			0.004	-0.002	-0.001
			(0.003)	(0.002)	(0.002)
hs__zip__inc__house__mean__decileD7			0.005	-0.001	-0.002
			(0.003)	(0.002)	(0.002)
hs__zip__inc__house__mean__decileD8			0.007	-0.003	-0.004
			(0.004)	(0.002)	(0.002)
hs__zip__inc__house__mean__decileD9			0.011*	-0.000	-0.002
			(0.005)	(0.003)	(0.003)
hs__zip__inc__house__mean__decileD10			0.040***	0.027***	0.023**
			(0.009)	(0.006)	(0.007)
hs__zip__pct__edu__baplus__all__decileD2			-0.000	0.000	-0.001
			(0.001)	(0.001)	(0.001)
hs__zip__pct__edu__baplus__all__decileD3			-0.001	-0.001	-0.002**
			(0.001)	(0.001)	(0.001)
hs__zip__pct__edu__baplus__all__decileD4			-0.000	-0.001	-0.003*
			(0.001)	(0.001)	(0.001)
hs__zip__pct__edu__baplus__all__decileD5			0.001	-0.000	-0.002
			(0.001)	(0.002)	(0.001)
hs__zip__pct__edu__baplus__all__decileD6			-0.001	-0.002	-0.004*
			(0.002)	(0.002)	(0.001)
hs__zip__pct__edu__baplus__all__decileD7			0.001	-0.000	-0.003
			(0.002)	(0.002)	(0.002)
hs__zip__pct__edu__baplus__all__decileD8			0.001	0.001	-0.003
			(0.003)	(0.002)	(0.002)
hs__zip__pct__edu__baplus__all__decileD9			0.009	0.008**	0.004
			(0.005)	(0.003)	(0.003)
hs__zip__pct__edu__baplus__all__decileD10			0.049***	0.048***	0.041***
			(0.009)	(0.008)	(0.007)
hs__zip__pct__pov__yes			0.000	0.000	0.000

(continued)

term	(1) Univ × State FE	(2) Univ × EPS FE	(3) Covar + Univ FE	(4) Covar + Univ × State FE	(5) Covar + Univ × EPS FE
			(0.000)	(0.000)	(0.000)
I(hs_zip_pct_pov_yes^2)			0.000	-0.000	-0.000
			(0.000)	(0.000)	(0.000)
hs_zip_pct_nhisp_black			0.000***	0.000***	0.000
			(0.000)	(0.000)	(0.000)
hs_zip_pct_nhisp_native			0.000	0.000	0.000
			(0.000)	(0.000)	(0.000)
hs_zip_pct_nhisp_asian			0.000	-0.000	-0.000
			(0.000)	(0.000)	(0.000)
hs_zip_pct_nhisp_nhpi			0.001	0.000	-0.000
			(0.000)	(0.000)	(0.000)
hs_zip_pct_nhisp_multi			0.001*	-0.000	-0.000
			(0.000)	(0.000)	(0.000)
hs_zip_pct_hisp_all			0.000*	0.000	0.000
			(0.000)	(0.000)	(0.000)
hs_univ_dist			-0.000***	-0.000**	-0.000*
			(0.000)	(0.000)	(0.000)
N	853,692	853,692	748,440	748,440	748,440
Adj. R^2	0.118	0.210	0.120	0.223	0.301
Within R^2	0.000	0.000	0.111	0.104	0.082
RMSE	0.143	0.134	0.149	0.140	0.132

Significance codes: *** p<0.001, ** p<0.01, * p<0.05

Table A2: Model fit statistics, fixed effects regression models of visits to high school i, separate models models by college j (public high schools only)

University	USNWR Rank	# PubHS Visits	AdjR2 State Only	AdjR2 EPS FE Only	AdjR2 Cov Only	AdjR2 Cov State	AdjR2 Cov EPS FE
Private Liberal Arts							
Williams	1	305	0.00	0.04	0.10	0.11	0.13
Swarthmore	3	441	0.02	0.09	0.10	0.12	0.17
Middlebury	9	376	0.03	0.08	0.12	0.14	0.18
Smith	15	404	0.03	0.10	0.11	0.12	0.17
Harvey Mudd	25	216	0.01	0.04	0.09	0.09	0.12
Colorado Coll.	25	300	0.05	0.09	0.12	0.16	0.19
Macalester	27	311	0.02	0.06	0.11	0.12	0.15
Scripps	28	169	0.02	0.06	0.07	0.08	0.13
Oberlin	36	502	0.02	0.09	0.20	0.22	0.25
Occidental	40	193	0.02	0.06	0.09	0.10	0.16
Sewanee	47	206	0.03	0.07	0.06	0.09	0.13
Conn Coll.	51	225	0.05	0.12	0.11	0.15	0.20
Private Research							
Northwestern	9	640	0.03	0.13	0.22	0.25	0.31
Notre Dame	19	422	0.01	0.07	0.16	0.18	0.21
Emory	21	315	0.02	0.11	0.13	0.15	0.22
Tufts	30	462	0.04	0.12	0.14	0.17	0.23
Boston Coll.	35	215	0.02	0.06	0.08	0.10	0.13
Tulane	41	503	0.03	0.13	0.17	0.20	0.26
Case Western Res.	42	227	0.01	0.06	0.11	0.12	0.16
Villanova	53	455	0.06	0.14	0.16	0.19	0.24
SMU	66	445	0.02	0.08	0.14	0.16	0.19
Baylor	76	580	0.08	0.15	0.14	0.20	0.24
U of Denver	80	413	0.11	0.16	0.12	0.21	0.26
TCU	80	509	0.02	0.11	0.16	0.18	0.24
Stevens Ins. Tech	80	310	0.12	0.20	0.09	0.17	0.24
Marquette	88	424	0.07	0.19	0.10	0.18	0.27
Public Research							
UC Berkeley	22	595	0.08	0.17	0.17	0.21	0.32
UC Irvine	35	436	0.09	0.16	0.12	0.17	0.25
UC San Diego	35	792	0.18	0.26	0.26	0.36	0.45
U of Georgia	47	514	0.16	0.22	0.11	0.25	0.30
U of Pitt	58	809	0.12	0.26	0.18	0.27	0.36
Rutgers	63	1065	0.25	0.40	0.22	0.37	0.48
UMass Amherst	66	768	0.24	0.35	0.19	0.37	0.43
UC Riverside	88	529	0.15	0.33	0.19	0.29	0.51
SUNY Stony Brook	88	842	0.19	0.45	0.19	0.29	0.49
CU Boulder	103	882	0.11	0.21	0.27	0.36	0.40
U of S.Carolina	118	1224	0.16	0.26	0.22	0.35	0.40
U of Kansas	124	966	0.24	0.31	0.15	0.36	0.41
UNL	133	1128	0.33	0.39	0.10	0.39	0.45
U of Alabama	143	2594	0.08	0.18	0.26	0.32	0.36
U of Cincinnati	143	926	0.22	0.36	0.12	0.31	0.42
U of Arkansas	160	731	0.19	0.25	0.09	0.26	0.31